

CEE PRO v2 Tutorial

A professional implementation guide for a TradingView strategy that detects contraction breakouts and manages long exits with ATR-based risk and exhaustion alerts.

PRACTICAL GUIDE

Prepared April 24, 2026 | Pine Script v6 | Strategy template

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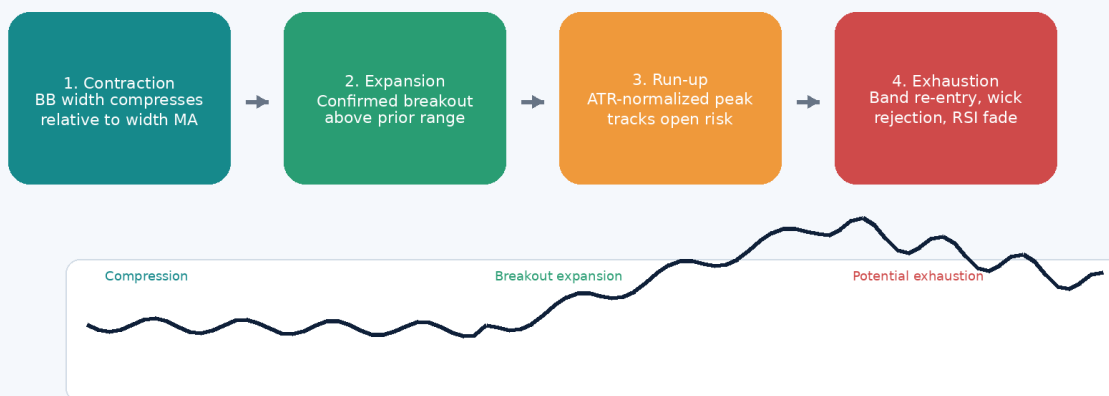
1. What was improved

The uploaded script already had a strong core: it identified Bollinger Band width contraction, entered long on an expansion breakout, trailed risk with ATR, and issued an exhaustion alert when the breakout showed rejection. CEE PRO v2 keeps that core and makes the strategy more production-ready, easier to tune, and easier to monitor.

- Grouped inputs for faster configuration and a cleaner TradingView settings panel.
- A transparent setup score that shows why a trade is armed or rejected.
- Optional filters for session, EMA slope, relative volume, ADX, VWAP, confirmed swing structure, and post-exit cooldown.
- A richer risk engine with ATR trail, optional breakeven migration, fixed ATR target, maximum-bars time exit, and two exhaustion responses: exit now or tighten stop only.
- Signal labels and a dashboard table that expose state, score, breakout status, trend status, RVOL, ADX, run-up, trail, and exhaustion score.
- Cleaner alert payloads designed for strategy alerts, webhook workflows, and manual trade review logs.

Important: the strategy is a technical template and does not guarantee profitability. Treat it as an engineering starting point, not a turnkey trading system.

Contraction -> Expansion -> Exhaustion lifecycle



2. Installation and first run

- 1 Download the Pine file named **Contraction_Expansion_Exhaustion_PRO_v2.pine**.
- 2 Open TradingView, create a new Pine Editor script, paste the full file, and save it.
- 3 Add it to the chart. Use a liquid symbol and the timeframe you actually intend to trade.
- 4 Open Strategy Tester. Confirm that commission, slippage, and position size match your intended execution assumptions.
- 5 Open the script settings. Start with optional filters disabled, then enable one filter at a time only if it improves out-of-sample behavior.
- 6 Create alerts only after the script is saved, added to the chart, and tested on the exact chart, symbol, and timeframe you will monitor.

Recommended first test:

Step	Action	Why it matters
Baseline	Run with default settings and no extra gates.	Confirms the original contraction-breakout idea is visible on your market.
Costs	Set realistic commission and slippage in Strategy Tester.	Stops a paper edge from disappearing in live execution.
Filters	Turn on volume, ADX, VWAP, or structure one at a time.	Prevents overfitting and shows which filter adds value.
Risk	Tune baseTrailATR, tightTrailATR, and breakeven trigger.	Exit behavior often matters as much as entry selection.
Forward test	Paper trade alerts for at least several market cycles.	Validates behavior on data the optimizer never saw.

3. Reading the lifecycle on the chart

The strategy is designed as a lifecycle. It does not buy simply because price is rising; it waits for compression, then expansion, then manages the later part of the move when momentum may be deteriorating.

Contraction

The background highlight appears when normalized Bollinger Band width is unusually narrow versus its own moving average and is at or near the lowest width over the contraction lookback. This is the setup state, not the entry by itself.

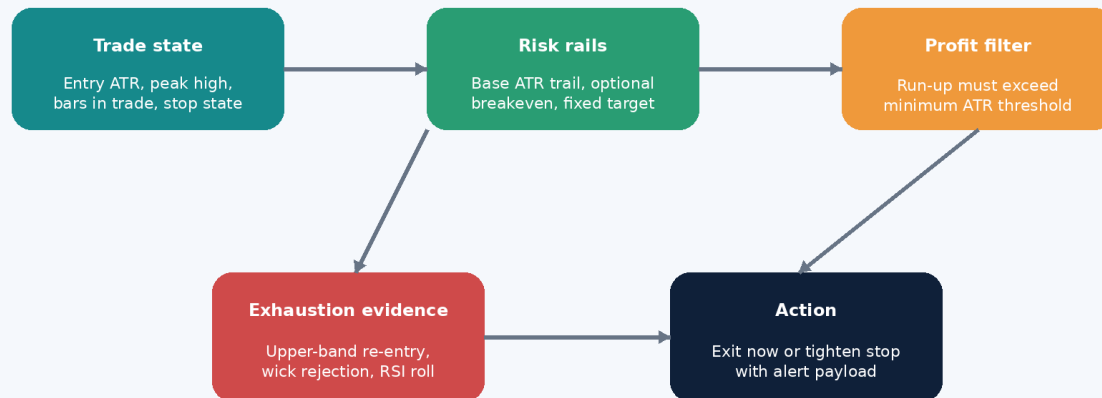
Expansion entry

A long entry requires a confirmed breakout above the prior range, a close above the upper band, a bar range larger than the ATR multiple, an acceptable close location, and any enabled quality gates. The dashboard score makes this pass/fail process visible.

Exhaustion management

After a profitable run-up, the script watches for band re-entry, upper-wick rejection, RSI rollover, and weak close location. When the exhaustion score is high enough, the script either exits immediately or tightens the trailing stop, depending on the selected exit mode.

Exit engine and alert paths



The strategy separates order-fill alerts from signal alerts. Order fills are emitted by `strategy.entry/exit/close` `alert_message` values; discretionary signal alerts come from `alert()` calls when enabled.

4. Input groups and recommended tuning sequence

Group	Key inputs	Practical guidance
Core indicators	EMA, Bollinger length/stdev, ATR, RSI	Begin with standard values. Change lengths only after you have observed whether the market is trend-sensitive or mean-reverting.
Contraction and breakout	contractionFactor, squeezeMemory, breakoutLen, expansionRangeATR, minCloseLocation, minSetupScore	Tune these first. They define the signal frequency and the quality of expansion bars.
Optional filters	Session, EMA slope, volume, ADX, VWAP, swing structure, cooldown	Use these as safeguards. Enable filters only when they remove low-quality trades without deleting the core edge.
Risk and exit	baseTrailATR, tightTrailATR, breakeven, fixed target, max bars	Control how long winners are allowed to breathe and how aggressively risk is reduced after run-up.
Alerts and visuals	alert() calls, dashboard, labels, bands, VWAP	Use visuals for development; hide labels and extra plots if you need a cleaner execution chart.

A safe tuning sequence is: contraction settings, then breakout range, then exit trail, then optional filters, then alerts. Avoid tuning all inputs simultaneously; that makes it difficult to know which change actually helped.

Core code excerpt

```
isContraction = bbwWidth < bbwWidthMa * contractionFactor and bbwWidth <= ta.lowest(bbwWidth, contractionLookback)
hadRecentContraction = not na(ta.barssince(isContraction)) and ta.barssince(isContraction) <= squeezeMemory
breakoutAboveRange = close > ta.highest(high, breakoutLen)[1]
expansionBar = close > upper and barRange > atr * expansionRangeATR and close > emaTrend
longEntry = barstate.isconfirmed and strategy.position_size == 0 and setupScore >= minSetupScore and expansionBar
```

5. Alerts, webhooks, and order-fill messages

CEE PRO v2 supports two alert paths. First, strategy order-fill messages are attached to strategy.entry(), strategy.exit(), and strategy.close(). Second, optional alert() calls can fire on entry and exit signals. When creating a TradingView alert, decide whether you want order fills, alert() calls, or both. Selecting both can be useful for audit logs, but it may create duplicate notifications if your automation endpoint treats both messages as executable orders.

Alert source	Message purpose	Typical use
strategy.entry alert_message	Reports long entry order fill context.	Automation or order-fill audit.
strategy.exit alert_message	Reports stop and optional target values.	Stop/target order monitoring.
strategy.close alert_message	Reports market close due to exhaustion or time exit.	Exit automation or manual review.
alert() calls	Reports signal events on the confirmed bar.	Webhook logs, discretionary alerts, or secondary notifications.

Suggested alert creation workflow

- 1 Add the saved strategy to the chart.
- 2 Open Create Alert and choose this strategy in the Condition dropdown.
- 3 Use order fills only for execution automation unless you intentionally want signal alerts too.
- 4 Paste a webhook URL only after testing messages with paper or sandbox routing.
- 5 Record every live alert during the first forward-test period and compare it to Strategy Tester order markers.

6. Testing workflow before real capital

A professional workflow separates idea validation from deployment. The purpose is to find out whether the strategy survives costs, market regimes, and unseen data, not simply whether it can be optimized to look attractive on one chart.

Professional validation workflow

	In-sample	Walk-forward	Forward test	Deployment
Metrics	Profit factor Expectancy Drawdown	Rolling windows Parameter drift Holdout data	Paper trades Live alert logs Slippage notes	Production caps Change log Kill switch
Robustness	Parameter grid Market regimes Trade count	Nested testing No retuning leak Stress spreads	Missed signals Webhook timing Data anomalies	Monthly review Retire rules Version control
Execution	Commissions Slippage Session rules	Gap behavior Stop fills Broker parity	Order routing Partial fills Latency	Size limits Broker errors Reconciliation
Governance	Hypothesis doc Risk budget Benchmark	Approval gates Review cadence Exception log	Incident notes Risk escalation Capital freeze	Audit trail User sign-off Archive

Metric	Why it matters	Caution
Net profit	Shows total historical result after assumptions.	Can be dominated by one outlier trade.
Profit factor	Compares gross profit to gross loss.	Can be inflated by small sample size.
Max drawdown	Measures adverse equity movement.	May understate risk if slippage or gaps are unrealistic.
Average trade	Shows whether the edge is large enough to survive fees.	Tiny averages are fragile.
Trade count	Shows whether results have enough observations.	Low counts are not robust evidence.

7. Customization recipes

For day trading

- Enable the session filter and match it to the instrument's most liquid hours.
- Consider relative volume and VWAP gates to avoid thin or directionless periods.
- Use tighter trailing stops only after testing their effect on whipsaws.

For swing trading

- Disable VWAP unless the chosen timeframe makes it meaningful for your process.
- Increase squeezeMemory or breakoutLen if contractions unfold over more bars.
- Consider a wider baseTrailATR and a maxBarsInTrade rule to avoid stale positions.

For automation

- Use order-fill alert messages for executable events.
- Keep alert() signal messages enabled only for logging or discretionary review unless your webhook parser can distinguish message types.
- Forward-test message timing, stop prices, and duplicate handling before routing to a broker.

8. Troubleshooting

Symptom	Likely cause	Fix
No trades	Filters too strict, minSetupScore too high, insufficient chart history, or VWAP/volume unavailable.	Disable optional gates, lower minSetupScore, and confirm BB/ATR values are not na.
Too many trades	Contraction threshold or breakout settings too loose.	Lower squeezeMemory, increase breakoutLen, raise expansionRangeATR, or enable structure/ADX filters.
Late exits	baseTrailATR too wide or exitMode set to tighten only.	Reduce baseTrailATR, lower minRunupATR, or use Exit now.
Frequent premature exits	trail settings too tight for the timeframe.	Increase baseTrailATR or breakEvenAtATR, and compare close-based vs intrabar behavior.
Alerts do not match expectations	Alert configured for only one trigger type or chart settings changed after alert creation.	Recreate the alert after saving the script and confirm Condition includes the intended trigger source.

9. Appendix: code architecture and references

The script uses confirmed bars for signal generation, strategy order functions for simulated orders, alert messages for fill reporting, and a floating table for state diagnostics. The delivered Pine file is the authoritative source; this tutorial explains its usage rather than replacing it.

References

- TradingView Pine Script v6 Language Reference Manual - <https://www.tradingview.com/pine-script-reference/v6/>
- TradingView Pine Script Strategies Concepts - <https://www.tradingview.com/pine-script-docs/concepts/strategies/>
- TradingView Pine Script Alerts Concepts - <https://www.tradingview.com/pine-script-docs/concepts/alerts/>
- TradingView Pine Script Bar States Concepts - <https://www.tradingview.com/pine-script-docs/concepts/bar-states/>
- TradingView Pine Script Tables Visuals - <https://www.tradingview.com/pine-script-docs/visuals/tables/>

Risk notice

Backtests are simulations. They may not reflect live liquidity, latency, rejected orders, gaps, partial fills, or data-feed differences. Nothing in this document is investment, tax, legal, or brokerage advice.